

# Lab Sheet 05

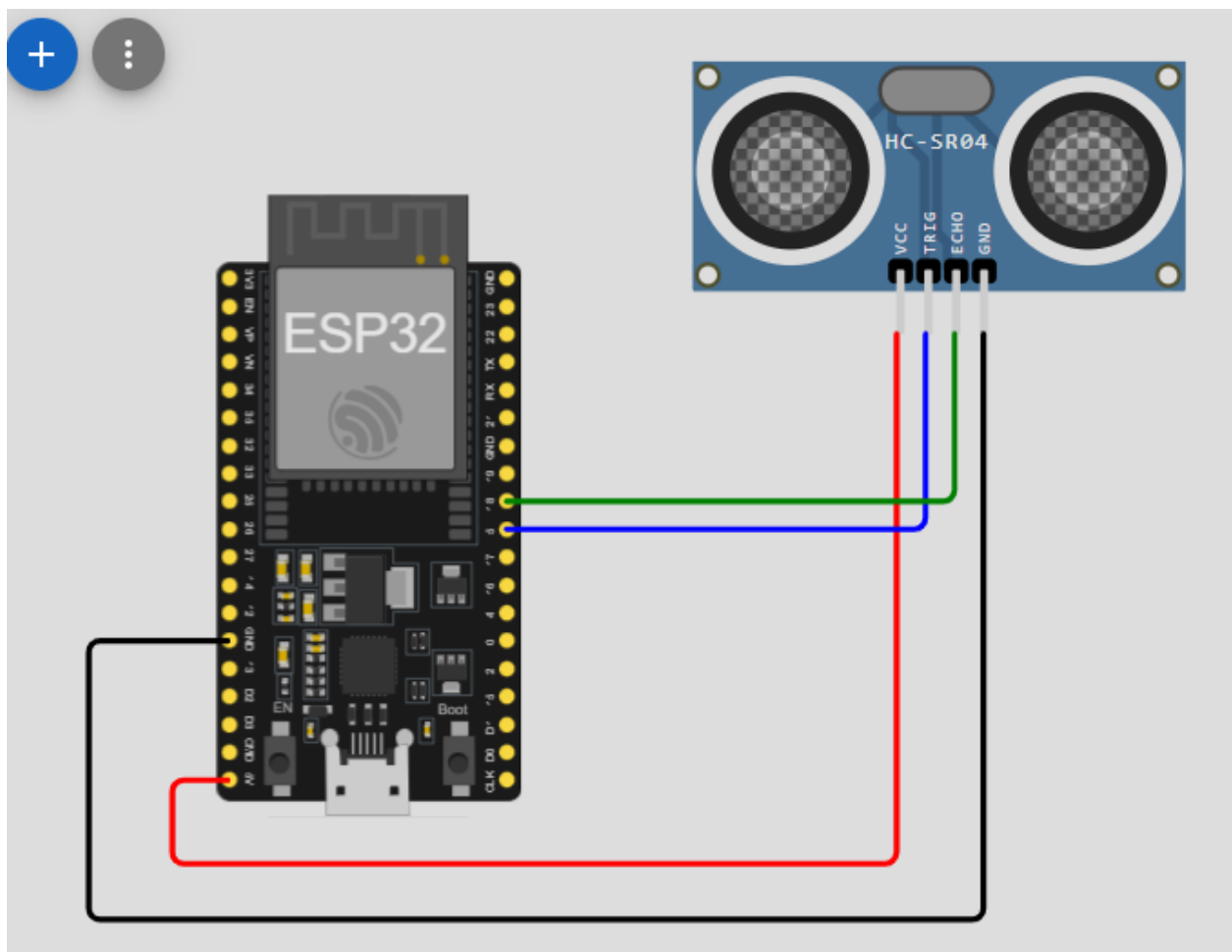
**Title:** Programming ESP32 with Sensors in Wokwi

**Aims:** To program the ESP32 to interface with various sensors, understand their libraries and communication protocols, and display real-time sensor data on the Serial Monitor.

**Tasks:**

## 1. HC-SR04 (Ultrasonic Distance Sensor)

- Write a program to measure distance in centimeters and display it on the Serial Monitor every second.



```
#include <Arduino.h>
const int trigPin = 5;
const int echoPin = 18;

void setup()
{
  Serial.begin(9600);
  Serial.println("\nHC-SR04 (Ultrasonic Distance Sensor) ");
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop()
{
  // Send pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

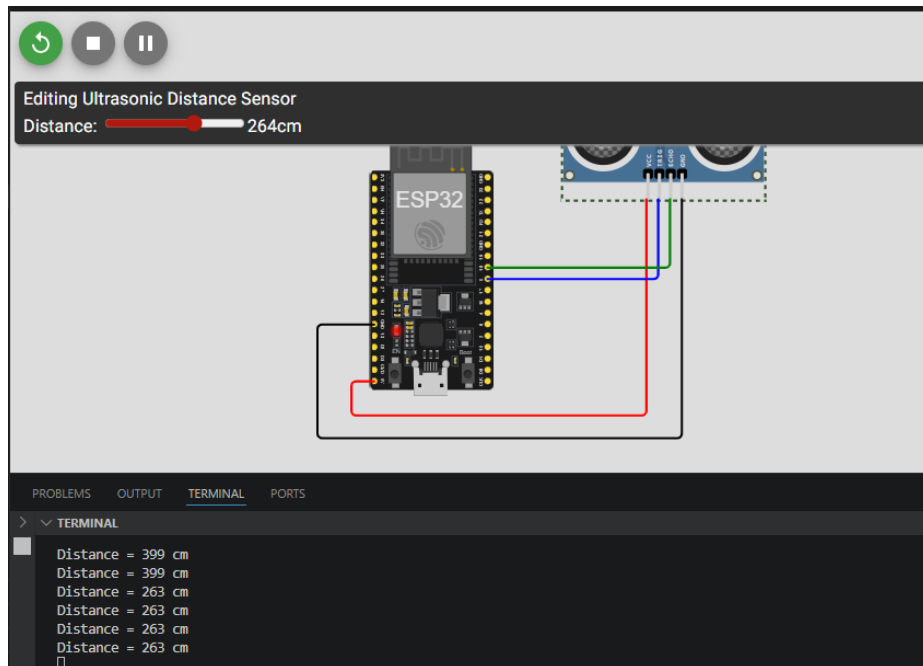
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read echo time
  long duration = pulseIn(echoPin, HIGH);

  // Calculate distance
  int distance = duration * 0.034 / 2;

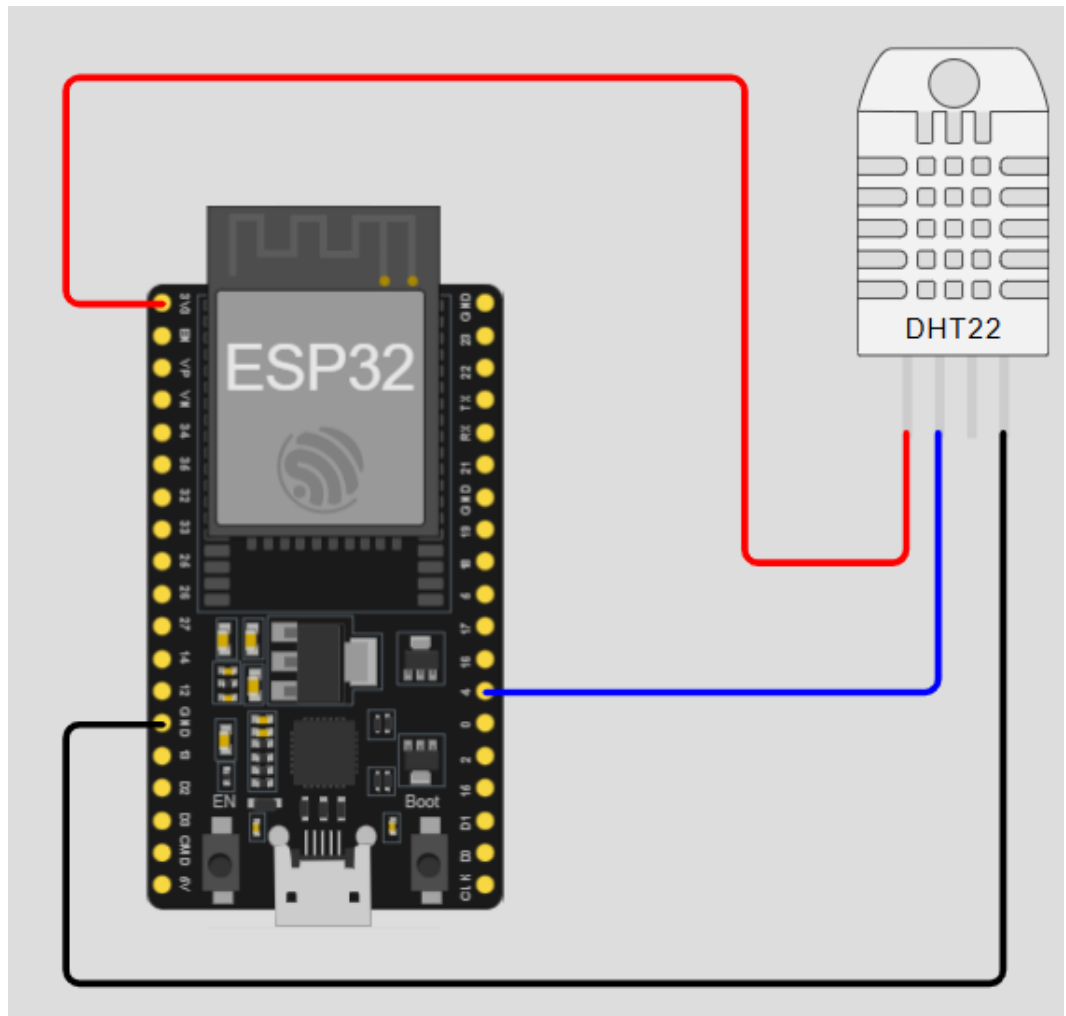
  // Print distance
  Serial.print("Distance = ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(1000);
}
```



## 2. DHT22 (Digital Humidity & Temperature Sensor)

- Write a program to read and display both temperature ( $^{\circ}\text{C}$ ) and humidity (%) every 2 seconds.



```

#include <Arduino.h>
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);

void setup()
{
  Serial.begin(115200);
  Serial.println("\nDHT22 (Digital Humidity & Temperature Sensor) ");
  dht.begin();
}

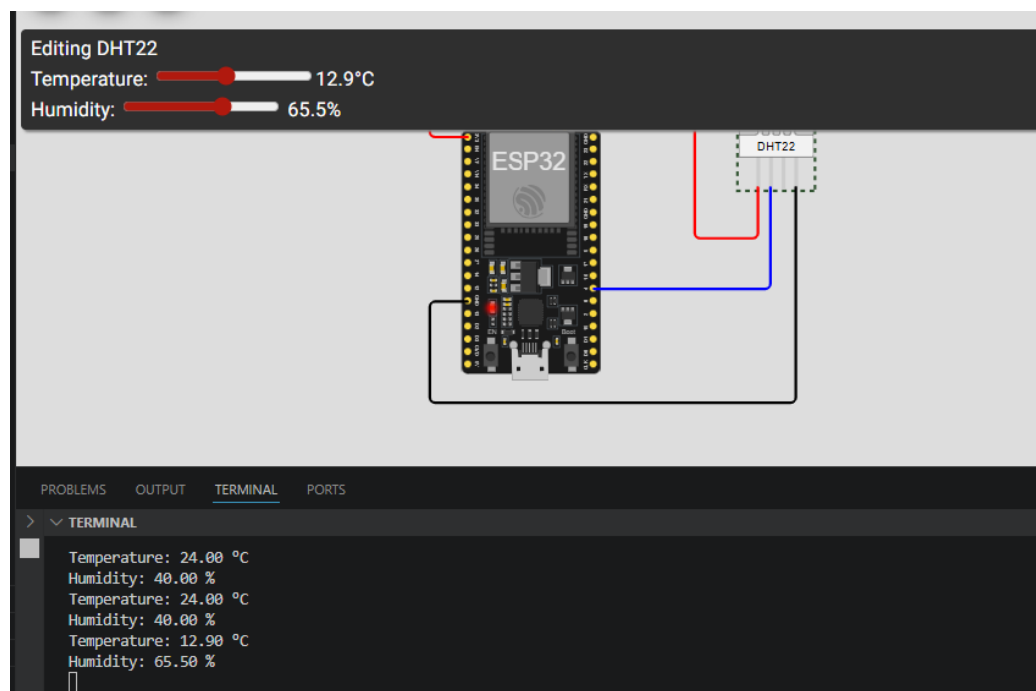
void loop()
{
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");

  Serial.print("Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");

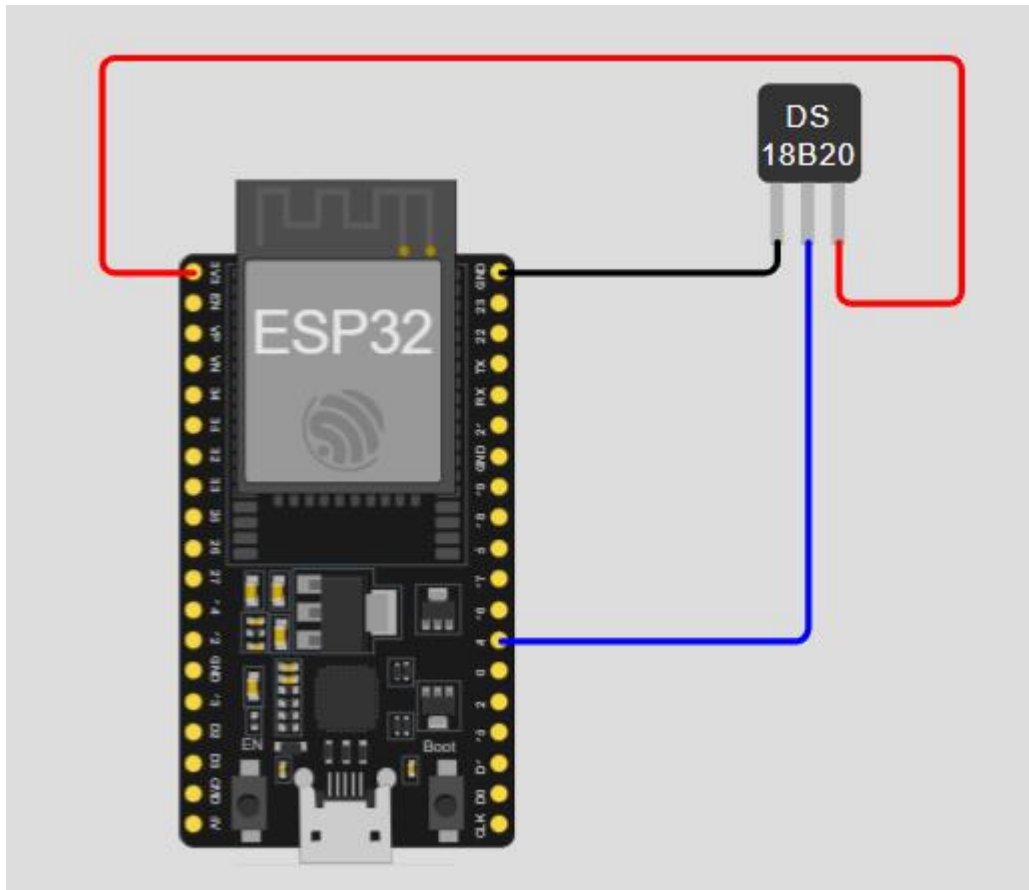
  delay(2000);
}

```



### 3. DS18B20 (One-Wire Digital Temperature Sensor)

- Write a program to read and display the temperature in Celsius.



```
#include <Arduino.h>
#include <OneWire.h>
#include <DallasTemperature.h>

#define ONE_WIRE_BUS 4

OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

void setup() {
  Serial.begin(9600);
  Serial.println("\nDS18B20 (One-Wire Digital Temperature Sensor) ");
  sensors.begin();
}

void loop() {
```

```

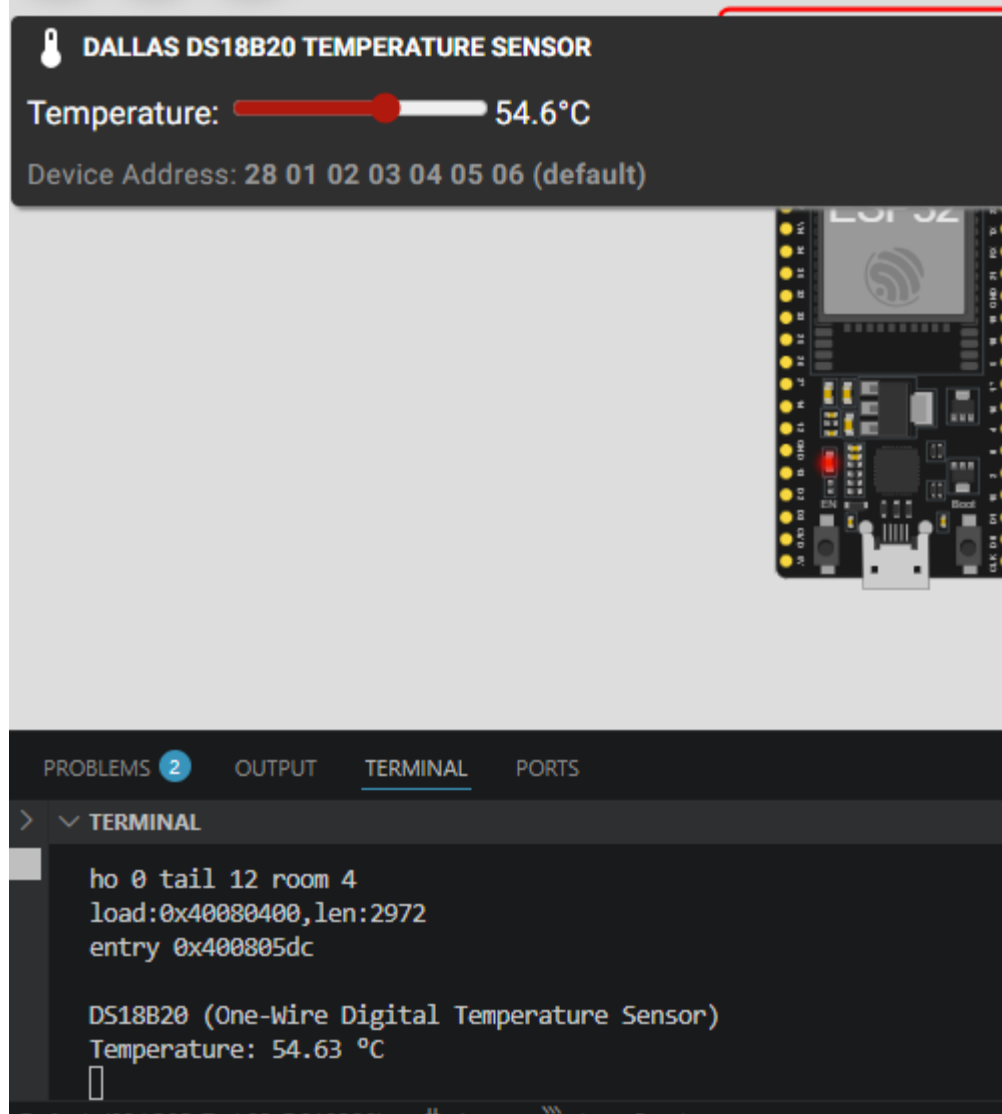
sensors.requestTemperatures();

float temp = sensors.getTempCByIndex(0);

Serial.print("Temperature: ");
Serial.print(temp);
Serial.println(" °C");

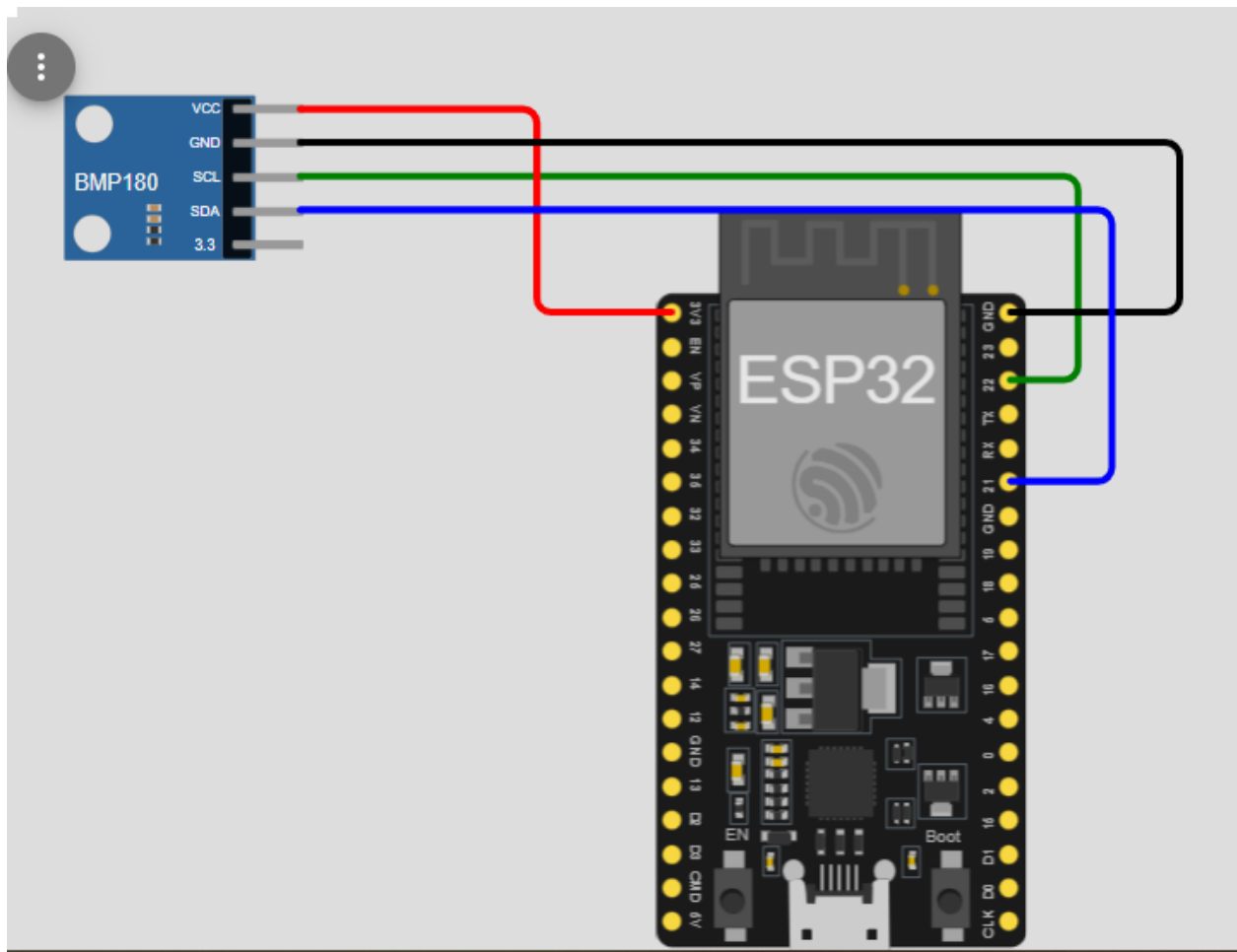
delay(1000);
}

```



#### 4. BMP180 (Barometric Pressure & Temperature Sensor)

- Write a program to display Temperature (°C), Pressure (hPa), and Altitude (meters).



```
#include <Arduino.h>
#include <Wire.h>
#include <Adafruit_BMP085.h>

Adafruit_BMP085 bmp;

void setup()
{
  Serial.begin(115200);
  Serial.println("\nBMP180 (Barometric Pressure & Temperature Sensor) ");
  bmp.begin();
}

void loop()
{
  float temperature = bmp.readTemperature();
  float pressure = bmp.readPressure() / 100.0;
```

```

float altitude = bmp.readAltitude();

Serial.print("Temperature: ");
Serial.print(temperature);
Serial.println(" °C");

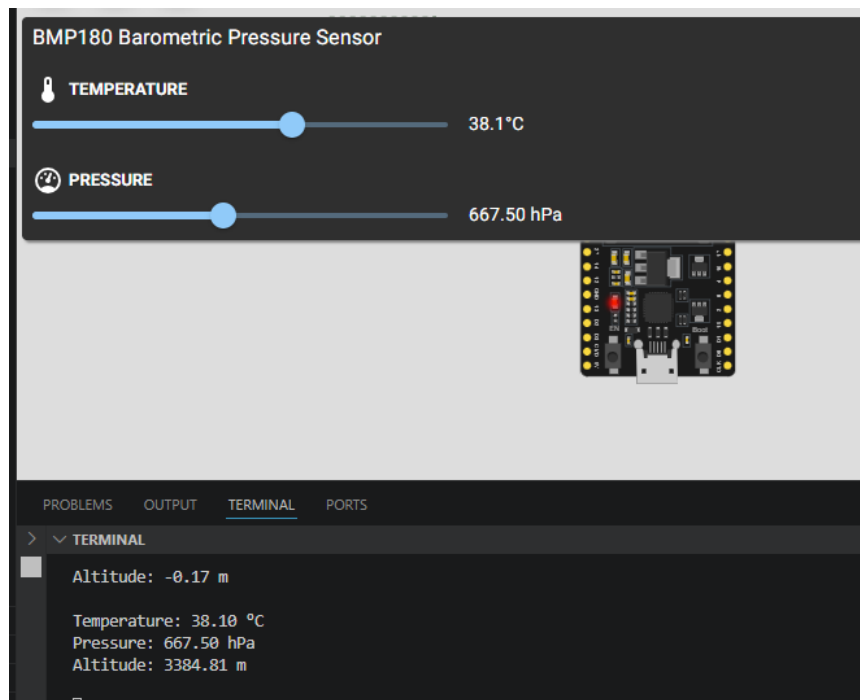
Serial.print("Pressure: ");
Serial.print(pressure);
Serial.println(" hPa");

Serial.print("Altitude: ");
Serial.print(altitude);
Serial.println(" m");

Serial.println();

delay(2000);
}

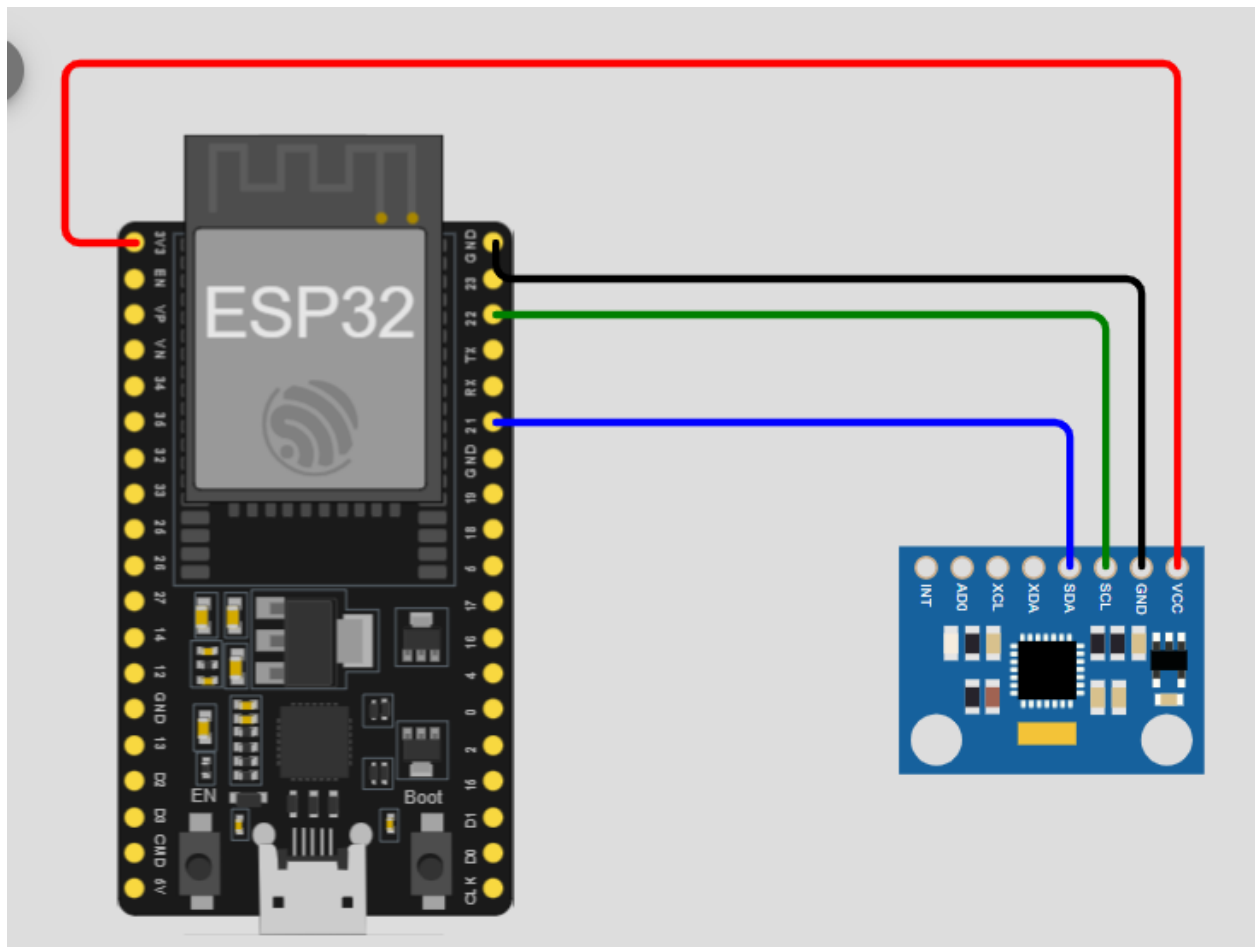
```



## 5. MPU6050 (3-Axis Accelerometer + Gyroscope + Temperature)

- Write a program to display acceleration (X, Y, Z) and gyroscope (X, Y, Z) values.





```
#include <Arduino.h>
#include <Wire.h>
#include <MPU6050.h>

MPU6050 mpu;

void setup() {
  Serial.begin(115200);
  Serial.println("\nMPU6050 (3-Axis Accelerometer + Gyroscope + Temperature) ");
  Wire.begin();
  mpu.initialize();
}

void loop() {

  int16_t ax, ay, az;
  int16_t gx, gy, gz;
```

```

// Read Accelerometer and Gyroscope values
mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);

Serial.println("Accelerometer");

Serial.print("X: ");
Serial.print(ax);

Serial.print("  Y: ");
Serial.print(ay);

Serial.print("  Z: ");
Serial.println(az);

Serial.println("Gyroscope");

Serial.print("X: ");
Serial.print(gx);

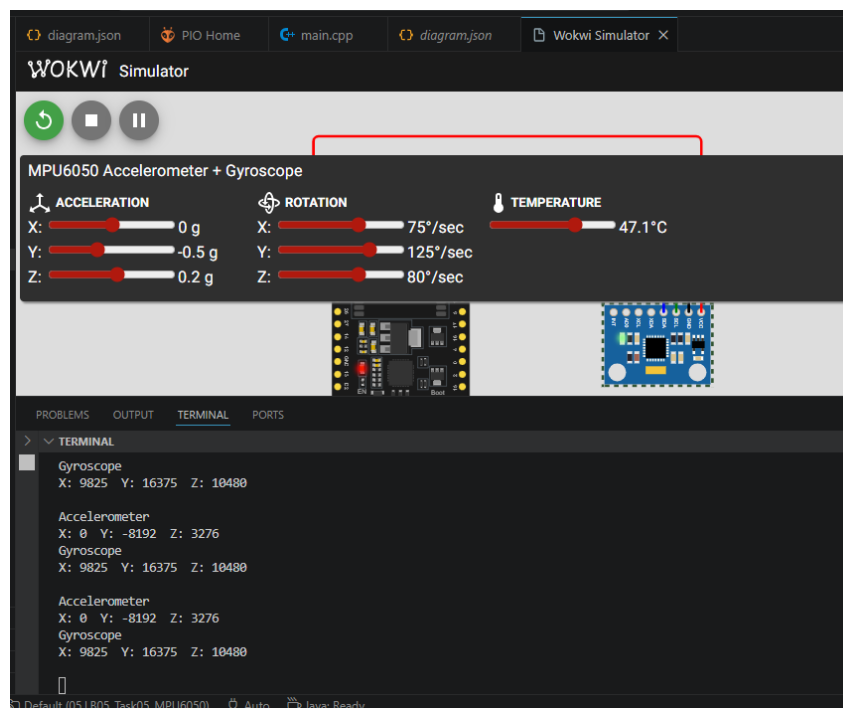
Serial.print("  Y: ");
Serial.print(gy);

Serial.print("  Z: ");
Serial.println(gz);

Serial.println();

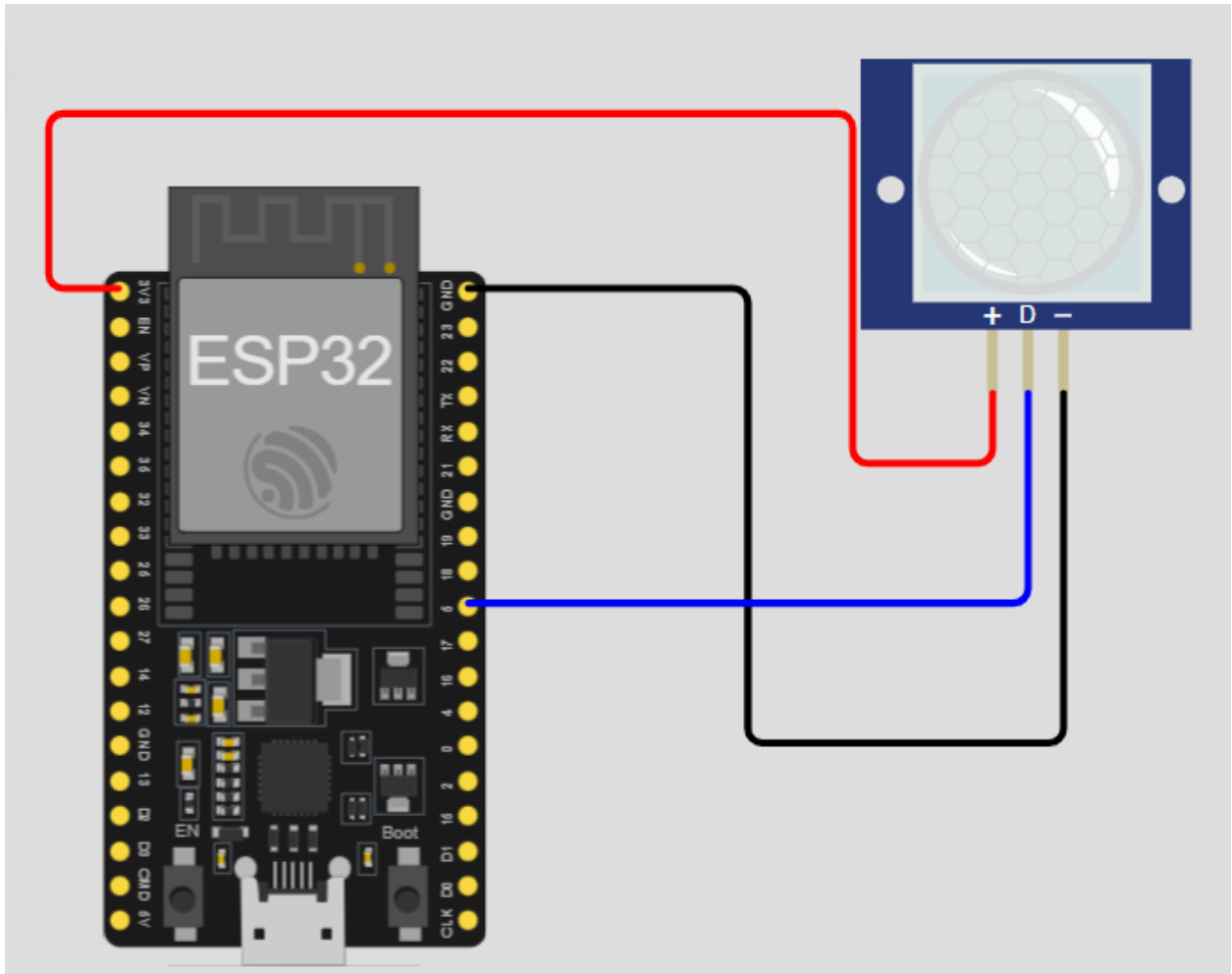
delay(1000);
}

```



## 6. PIR Motion Sensor (Passive Infrared)

- Write a program to detect motion and print "Motion Detected" or "No Motion" continuously.



```
#include <Arduino.h>
const int pirPin = 5;

void setup()
{
  Serial.begin(115200);
  Serial.println("\nPIR Motion Sensor (Passive Infrared) ");
  pinMode(pirPin, INPUT);
}

void loop()
{
```

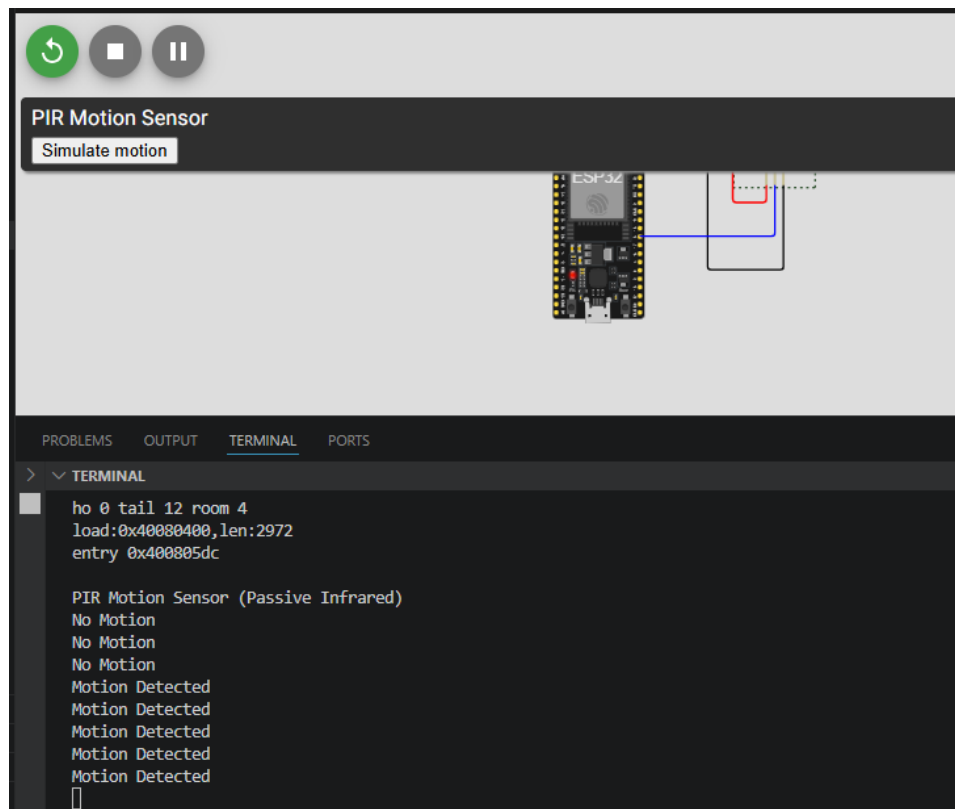
```

int motion = digitalRead(pirPin);

if (motion == HIGH)
{
    Serial.println("Motion Detected");
}
else
{
    Serial.println("No Motion");
}

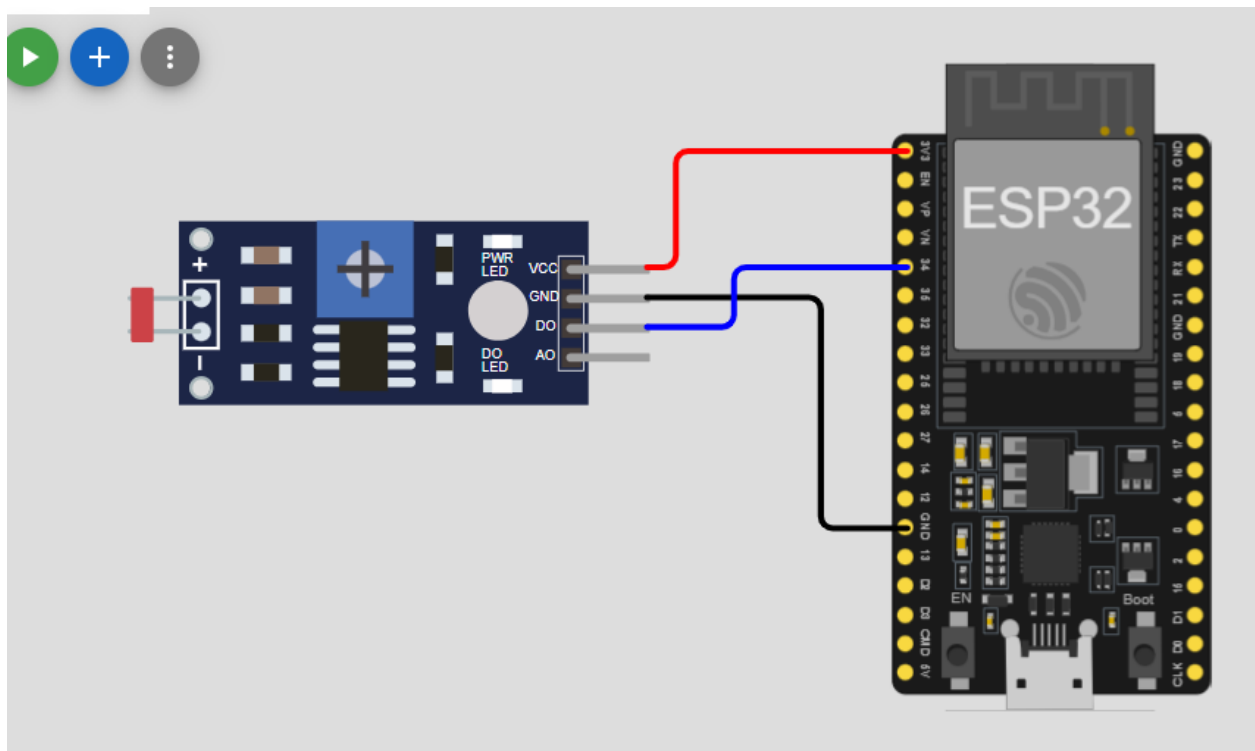
delay(1000);
}

```



## 7. Photoresistor (LDR) Light Sensor

- Write a program to read and display the analog light intensity value.



```
#include <Arduino.h>
const int LdrPin = 34;

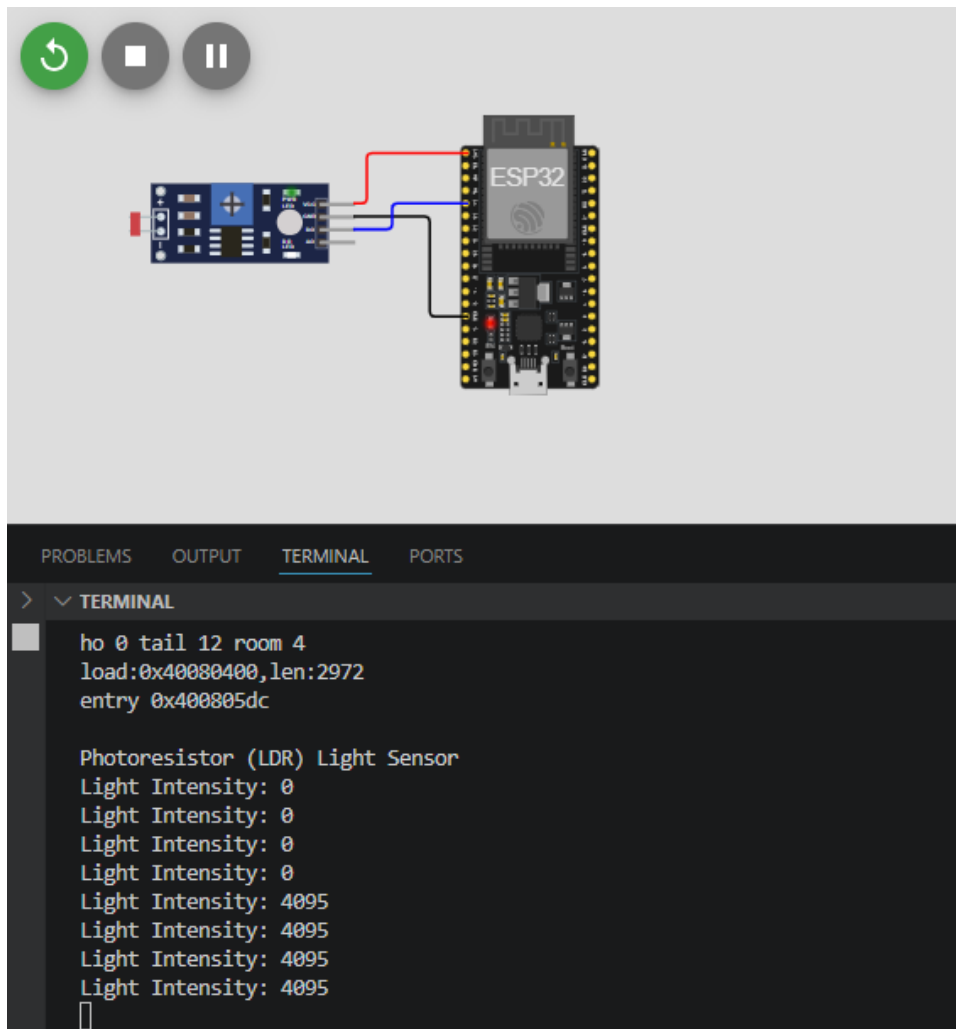
void setup() {
  Serial.begin(115200);
  Serial.println("\nPhotoresistor (LDR) Light Sensor ");
}

void loop() {

  int LightValue = analogRead(LdrPin);

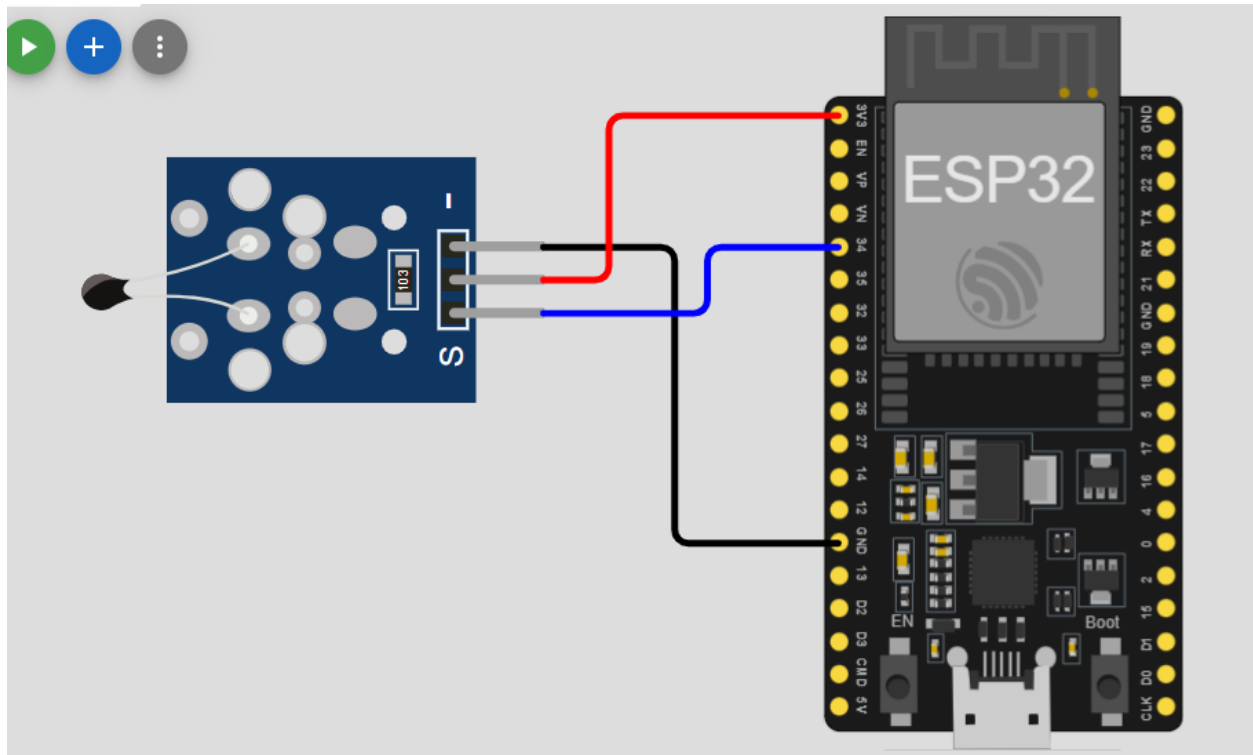
  Serial.print("Light Intensity: ");
  Serial.println(LightValue);

  delay(1000);
}
```



## 8. Analog Temperature Sensor (NTC Thermistor)

- Write a program to calculate and display temperature in °C using the thermistor.



```
#include <Arduino.h>
const int thermistorPin = 34;

void setup() {
  Serial.begin(115200);
  Serial.println("\nAnalog Temperature Sensor (NTC Thermistor) ");
}

void loop() {

  int value = analogRead(thermistorPin);

  // Convert analog value to temperature
  float temperature = map(value, 0, 1023, -20, 100);

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");

  delay(1000);
}
```





```

const int mq2Pin = 34;

void setup() {
  Serial.begin(115200);
  Serial.println("MQ2_Gas_Sensor ");
}

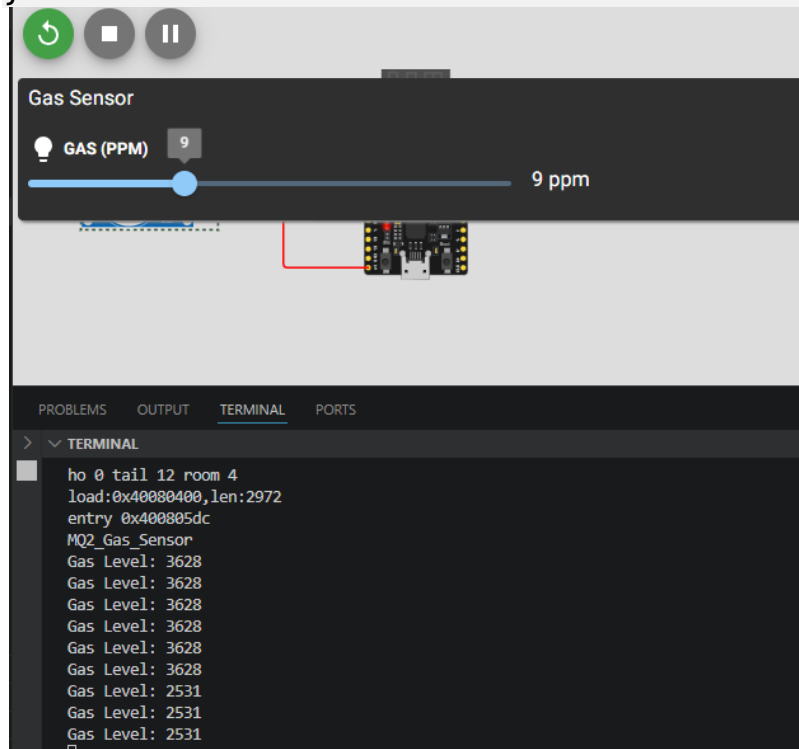
void loop() {

  int gasValue = analogRead(mq2Pin);

  Serial.print("Gas Level: ");
  Serial.println(gasValue);

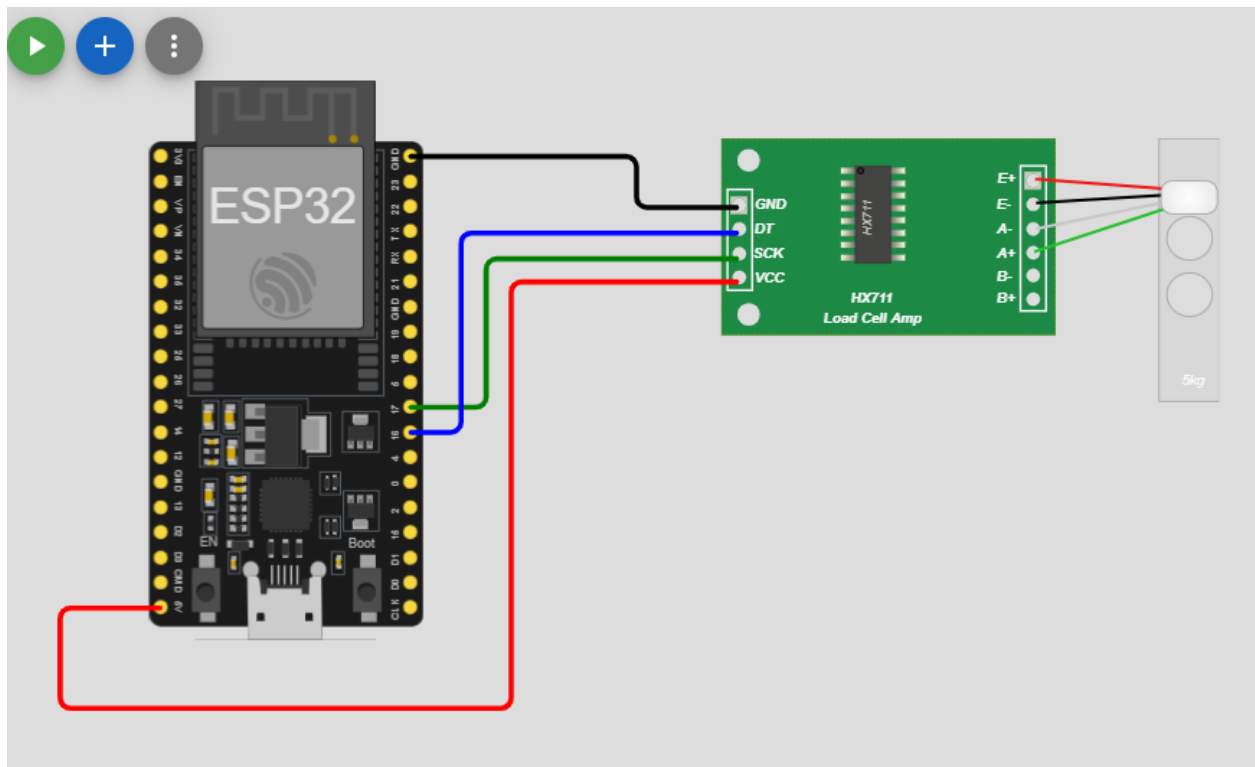
  delay(500);
}

```



## 10. HX711 Load Cell Amplifier

- Write a program to display weight in grams. Include taring (zeroing) at startup.



```
#include <Arduino.h>
#include "HX711.h"

#define DT 16
#define SCK 17

HX711 scale;

void setup()
{
  Serial.begin(115200);
  Serial.println("\nHX711 Load Cell Amplifier ");
  scale.begin(DT, SCK);

  // Tare (zero the scale at startup)
  Serial.println("Taring... remove any weight");
  delay(2000);
  scale.tare();

  Serial.println("Ready");
}

void loop()
```

```

{

  // Set calibration factor (you MUST adjust this for your load cell)
  float calibration_factor = -7050;

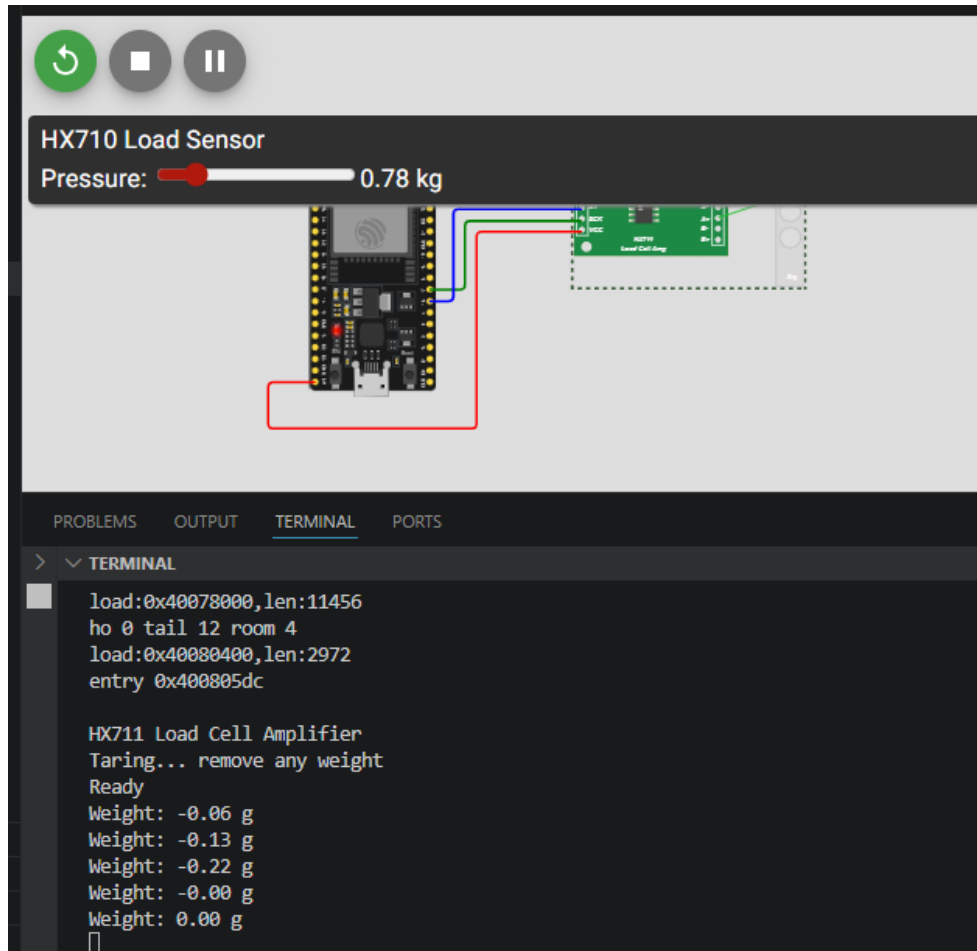
  scale.set_scale(calibration_factor);

  float weight = scale.get_units(10); // average of 10 readings

  Serial.print("Weight: ");
  Serial.print(weight);
  Serial.println(" g");

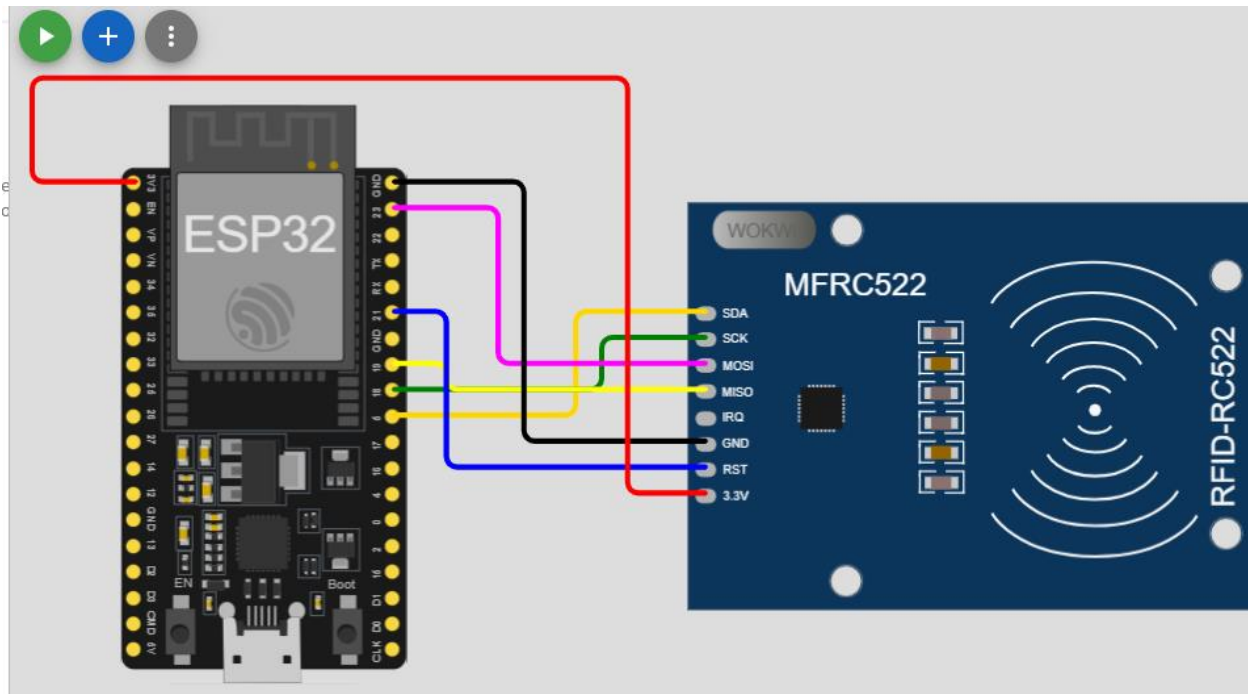
  delay(500);
}

```



## 11. MFRC522 RFID/NFC Reader

- Write a program to read an RFID card and print its UID on the Serial Monitor.



```
#include <Arduino.h>
#include <SPI.h>
#include <MFRC522.h>

#define SS_PIN 5
#define RST_PIN 22

MFRC522 rfid(SS_PIN, RST_PIN);

void setup() {
  Serial.begin(115200);
  Serial.println("\nMFRC522 RFID/NFC Reader ");
  SPI.begin();
  rfid.PCD_Init();

  Serial.println("Place RFID card near reader...");
}

void loop() {

  if (!rfid.PICC_IsNewCardPresent()) {
    return;
  }
}
```

```

}

if (!rfid.PICC_ReadCardSerial()) {
    return;
}

Serial.print("UID: ");

for (byte i = 0; i < rfid.uid.size; i++) {
    Serial.print(rfid.uid.uidByte[i] < 0x10 ? " 0" : " ");
    Serial.print(rfid.uid.uidByte[i], HEX);
}

Serial.println();

rfid.PICC_HaltA();
}

```



## MFRC522 RFID Reader

Card

 Yellow Card ▾

TAP



Hold

UID: 55:66:77:88



PROBLEMS OUTPUT TERMINAL PORTS

> ▾ TERMINAL

```

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1156
load:0x40078000,len:11456
ho 0 tail 12 room 4
load:0x40080400,len:2972
entry 0x400805dc

```

```

MFRC522 RFID/NFC Reader
Place RFID card near reader...
UID: 55 66 77 88

```

